Exact big Ramsey degrees of finite powers of omega by canonical coefficient patterns

Joanna Boyland, William Gasarch, Nathan Hurtig, Robert Rust

arxiv:2305.07192v3; CC BY 4.0

Editorial note. The original source is retained. In the displayed construction at lines1053-1058, S_0 is not defined and the S_i labels are shifted; the preceding definition at1052 and the explicit inequality a_(2i-1)<min A_i at1074-1075 specify the intended pairing of S_i with a_(2i-1), for i>=1. This indexing observation requires independent confirmation; the extraction does not silently replace the display.

Editorial note. The source uses one-based element labels even though [n] was introduced with zero-based notation. These labels index an unordered n-element edge. No claim is counted about the independent recurrence for the numbers of patterns.

Editorial scope. One counted general equality theorem, with both bounds and its full bespoke copy-construction lemma included as solution obligations. The previously published n=2,d=2 example is only an illustration. Infinite Ramsey and elementary finite ordinal arithmetic are explicitly allowed standard inputs; the separate pattern-count recurrence is outside the counted task.

Allowed standard prerequisites (editorial).

- Infinite Ramsey theorem for finite colorings of fixed-size subsets of a countable set. For each finite arity and finite number of colors, an infinite homogeneous subset exists.. source0.tex:241-251. Applied once to the finite vector of colors attached to all canonical coefficient patterns.
- Elementary arithmetic of finite powers of omega. A subset of omega^(j+1) of full order type has infinitely many omega^j rows with full order type; otherwise its order type is smaller.. Choose successively arbitrarily large coefficient values while retaining the full residual row type in the lower-bound proof.

Definition 0.1. Let $\mathbb{N}$ be the set of natural numbers, including 0. Let $n, c \in \mathbb{N}$ and S be a set. Then we define $[c]$ to be $\{0, \ldots, c-1\}$ (note $0 \in [c]$, but $c \notin [c]$). Note that if $c = 0$ then $[c] = \emptyset$. We use $\binom{S}{n}$ to denote the set of all n-element subsets of S.

Definition 0.2. A function $\mathrm{COL}\colon \binom{S}{n} \to [c]$ is called a *coloring*. A set $H \subseteq S$ is called *homogeneous* if COL restricted to $\binom{H}{n}$ is constant, that is, $\left|\mathrm{COL}\left(\binom{H}{n}\right)\right| \leq 1$.

Definition 0.3. Let $\mathcal{A} = (A, \preceq_A)$ and $\mathcal{B} = (B, \preceq_B)$ be ordered sets, where $\preceq_A, \preceq_B$ are total orders on A, B respectively.

1. Two ordered sets $\mathcal{A}, \mathcal{B}$ are *order-equivalent*, denoted $\mathcal{A} \approx \mathcal{B}$, if there exists an order-preserving bijection $f\colon A \to B$ such that, for all $a_1, a_2 \in A$:

$$a_1 \preceq_A a_2 \iff f(a_1) \preceq_B f(a_2).$$

2. The ordered set $\mathcal{A}+\mathcal{B}$ is defined to be $(A \sqcup B, \preceq)$, where the disjoint union $A \sqcup B$ is formally defined to be $(\{0\} \times A) \cup (\{1\} \times B)$, and where $\preceq$ is the dictionary ordering on $A \sqcup B$. Note that $+$ agrees with the definition of ordinal addition, and is not commutative in general.

3. Let $\mathcal{A} = (A, \preceq_A)$ be an ordered set with $A \subseteq \mathbb{Z}$. Then define $-\mathcal{A}$ to be $(-A, \preceq_{-A})$, where $-A = \{-a\colon a \in A\}$, and $-a \preceq_{-A} -b \iff b \preceq_A a$.

Throughout this paper, we conflate the notation for an ordered set and its underlying set. For example, if A is an ordered set, we might still use the notation $\binom{A}{2}$.

Definition 0.4. Let S be an ordered set, $S' \subseteq S$, and $n, c, t \in \mathbb{N}$. Let $\mathrm{COL}\colon \binom{S}{n} \to [c]$ be a coloring.

1. S' is *homogeneous* if $\left|\mathrm{COL}\left(\binom{S'}{n}\right)\right| = 1$.

2. S' is *t-homogeneous* if $\left|\mathrm{COL}\left(\binom{S'}{n}\right)\right| \leq t$.

3. S' is *S-t-homogeneous* if both S' is t-homogeneous and $S' \approx S$.

Definition 0.5. We identify every ordinal α as the ordered set of all ordinals $\beta < \alpha$, with 0 as the least ordinal. Additionally, let ω be the ordered set containing only the naturals, ζ be the ordered set containing only the integers, and η be the ordered set containing only the rationals, all under their respective natural orderings.

Definition 0.6. Let S be an ordered set. For any $n \in \mathbb{N}$, $T(n, S)$ is the least $t \in \mathbb{N}$ such that for all constants $c \geq 1$ and colorings $\mathrm{COL}\colon \binom{S}{n} \to [c]$, there exists some $S' \subseteq S$ such that S' is S-t-homogeneous. Note that t is independent of c. If no such t exists then we define $T(n, S) = \infty$. The quantity $T(n, S)$ is called the *big Ramsey degree* of n in S. The term was first coined by Kechris et al. [2]. Other sources use the Ramsey arrow notation modified for ordered sets. In that notation, $T(n, S) = t$ is short-hand for both $S \to (S)^n_{c,t}$ and $S \not\to (S)^n_{T(n,S),t-1}$ for all $c \in \mathbb{N}$. We use the $T(n, S)$ notation in this paper.

Theorem 0.7. *For all $n \in \mathbb{N}$, $T(n, \omega) = 1$.*

Definition 0.8. If $e \in \binom{\omega^d}{n}$, we refer to e as an *edge*. Then e has n *elements* selected from ω^d. Hence each element of e is some ordinal number less than ω^d. That element can be represented as a polynomial of degree $< d$ in ω. We refer to the natural-valued coefficients of these polynomials as the *coefficients* of e.

Theorem 0.9. *The big Ramsey degree $T(2, \omega^2) = 4$.*

We introduce a concept called *coloring rules* (hereafter CRs) to prove general results about $T(n, \omega^d \cdot k)$. The concept behind CRs is built on the ideas of Blass et al. [1].

Definition 0.10. We now define CRs (coloring rules) rigorously. We impose a structure on edges and list criteria that CRs must satisfy. For $n, d, k \in \mathbb{N}$, an edge $e = \{p_1, \ldots, p_n\}$ in $\binom{\omega^d \cdot k}{n}$ consists of n elements of $\omega^d \cdot k$. Each element p_i is equal to

$$\omega^d \cdot b_i + \omega^{d-1} \cdot a_{i,d-1} + \omega^{d-2} \cdot a_{i,d-2} + \cdots + \omega^1 \cdot a_{i,1} + a_{i,0},$$

where $a_{i,j} \geq 0$ and $0 \leq b_i < k$.

Thus, any edge e is defined by the n values of the b_i and the $n \cdot d$ values of the $a_{i,j}$.

A *CR (coloring rule)* on $\binom{\omega^d \cdot k}{n}$ is a pair $(\overline{B}, \preceq_{\overline{A}})$ of constraints on these values b_i and $a_{i,j}$ satisfying certain properties we enumerate below. Firstly, $\overline{B}$ is an assignment of the values for the b_i; formally, it is a map $\overline{B} : [n] \to \{0, \ldots, k-1\}$ from indices of the b_i to the values of the b_i. Having $\overline{B}(i) = v$ means we constrain $b_i = v$. Meanwhile, $\preceq_{\overline{A}}$ is a total preorder (i.e. every two of elements are comparable, and $\preceq_{\overline{A}}$ is reflexive and transitive) on the indices of the $a_{i,j}$. Having $(i,j) \preceq_{\overline{A}} (i',j')$ means we constrain $a_{i,j} \leq a_{i',j'}$.

We often denote $\overline{B}$ using an ordered list of clauses:

$$b_1 = \overline{B}(1), b_2 = \overline{B}(2), \ldots, b_n = \overline{B}(n).$$

We often denote $\preceq_{\overline{A}}$ by a permutation of the expressions $a_{1,0}, \ldots, a_{n,d-1}$ interspersed by either $<$ or $=$. We write $a_{i,j} < a_{i',j'}$ to mean that both $(i,j) \preceq_{\overline{A}} (i',j')$ and $(i',j') \not\preceq_{\overline{A}} (i,j)$ and we write $a_{i,j} = a_{i',j'}$ to mean both $(i,j) \preceq_{\overline{A}} (i',j')$ and $(i',j') \preceq_{\overline{A}} (i,j)$. Note that $x = y < z$ and $y = x < z$ are two representations of the same preorder, so this notation is not unique. One example of such a representation is

$$a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0}.$$

We refer to the expressions $a_{i,j}$ and b_i when used in our notation for $\overline{B}$ and $\preceq_{\overline{A}}$ as variables, and will describe different preorders $\preceq_{\overline{A}}$ by discussing permutations of some sequence of $a_{i,j}$.

To be a CR the following must hold:

1. If $d \geq 1$, then for all $i < i'$, $a_{i,0} < a_{i',0}$. Otherwise when $d = 0$, $b_i < b_{i'}$ for all $i < i'$. (The element indices are ordered by their lowest-exponent variable.)

2. For all i, i', $(\exists j \ a_{i,j} = a_{i',j}) \implies b_i = b_{i'}$. (Two elements with any a values that are the same must have the same b value.)

3. For all $j > j'$, $a_{i,j} < a_{i,j'}$. (The high-exponent variables of each element are strictly less than the low-exponent variables.)

4. $a_{i,j} = a_{i',j'} \implies j = j'$. (Only variables with the same exponent can be equal.)

5. $a_{i,j} \neq a_{i',j} \implies a_{i,j-1} \neq a_{i',j-1}$ for all $j > 0$. (Two elements that differ in a high-exponent variable differ in all lower-exponent variables.)

An example of a CR for $\binom{\omega^2}{2}$ is

$$b_1 = 0,\ b_2 = 0,\ a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0}.$$

Note that because $k = 1$ in the example, we must have $b_i = 0$ for every i.

Definition 0.11. Two CRs are *equivalent* whenever their constraints $\overline{B}, \preceq_{\overline{A}}$ are exactly equal.

Definition 0.12. The *size* of a CR is how many equivalence classes its $a_{i,j}$ form under $\preceq_{\overline{A}}$: for example, $a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0}$ would have size $p = 3$ regardless of b_i. Therefore a CR's size p can be no larger than $n \cdot d$.

Definition 0.13.

1. $P_p\left(n, \omega^d \cdot k\right)$ is the number of CRs of size p there are for $\binom{\omega^d \cdot k}{n}$.
2. $P\left(n, \omega^d \cdot k\right)$ is the total number of CRs there are for $\binom{\omega^d \cdot k}{n}$ of any size. It can be calculated as
$$\sum_{p=0}^{n \cdot d} P_p(n, \omega^d \cdot k).$$

Eventually, we will show $T(n, \omega^d \cdot k) = P(n, \omega^d \cdot k)$.

Definition 0.14. An edge

$$\{\omega^d \cdot b_i + \omega^{d-1} \cdot a_{i,d-1} + \omega^{d-2} \cdot a_{i,d-2} + \cdots + \omega^1 \cdot a_{i,1} + a_{i,0} \colon 1 \leq i \leq n\}$$

satisfies some CR if $\overline{B}(i) = b_i$ for every $1 \leq i \leq n$ and if $(i,j) \preceq_{\overline{A}} (i',j') \iff a_{i,j} \leq a_{i',j'}$ for all $1 \leq i, i' \leq n$ and $0 \leq j, j' < d$. Note that while every edge describes some map $\overline{B}$ and order $\preceq_{\overline{A}}$, if those constructs do not meet the requirements in Definition 0.10, that edge does not satisfy any CR. We extend and formalize the idea of coefficients introduced in Definition 0.8 to be the set of $a_{i,j}$ terms.

1 Big Ramsey Degrees of ω^d

This section is devoted to the case where $k = 1$ in $\binom{\omega^d \cdot k}{n}$. When $k = 1$, each b_i in a CR is forced to be 0. Then all b_i values are the same, so criterion 2 of Definition 0.10 is always satisfied. In this section, our proofs focus only on how $\preceq_{\overline{A}}$ permutes the $a_{i,j}$ variables. When values for b_i are not specified, they are assumed to be all 0 by default.

Theorem 1.1. *For $n, d \in \mathbb{N}$, $T\left(n, \omega^d\right) \geq P\left(n, \omega^d\right)$. Therefore by Theorem 1.3, $T(n, \omega^d) = P(n, \omega^d)$.*

Lemma 1.2. *For $n, d \in \mathbb{N}$ and some ordered set $G \subseteq \omega$ with $G \approx \omega$, there exists some $H \subseteq \omega^d$ with $H \approx \omega^d$ where for all $e \in \binom{H}{n}$, e satisfies a CR of $\binom{\omega^d}{n}$ and each coefficient of e (see Definition 0.14) is contained in G.*

Proof. Let $G = \{a_1 < a_2 < a_3 < \cdots\}$ with each $a_i \in \omega$ and $G \approx \omega$. We proceed by induction on d.

When $d = 0$, $\omega^0 \approx 1$ and so $H = \{0\} = 1$ suffices. When $n < 2$, the only CR of $\binom{H}{n} = \binom{1}{n}$ is satisfied by every edge of $\binom{1}{n}$ and has $n \cdot d = 0$ coefficients: the 1 in $\omega^0 \cdot 1$ is a b_i term, not an $a_{i,j}$. When $n \geq 2$, there are no CRs of $\binom{1}{n}$ so the result is vacuously satisfied.

When $d = 1$, $\omega^1 \approx \omega$ so $H = G$ suffices, as the single CR for $\binom{\omega}{n}$ is satisfied by all n-subsets of ω, and their n coefficients are in H, hence in G.

For $d \geq 2$, partition G into infinite sets, each order-equivalent to ω:

$$\begin{aligned} A_0 &= \{a_1, a_3, a_5, \ldots\} \\ A_1 &= \{a_2, a_6, a_{10}, \ldots\} \\ A_2 &= \{a_4, a_{12}, a_{20}, \ldots\} \\ A_3 &= \{a_8, a_{24}, a_{40}, \ldots\} \\ &\vdots \end{aligned}$$

Formally,

$$A_i = \{a_j \colon j = 2^i + k2^{i+1}, k \in \mathbb{N}\}.$$

Reserve A_0 for later. Apply the inductive hypothesis, taking G to be A_i, for all $i \geq 1$. This yields $S_i \approx \omega^{d-1}$ for all $i \geq 1$. Then for all $i \geq 1$, for all $e \in \binom{S_i}{n}$, e satisfies a CR of $\binom{\omega^{d-1}}{n}$ and each coefficient of e is contained in A_i. We now pair each element of A_0 with an S_i. Let

$$\begin{aligned} H = \quad & \{\omega^{d-1}a_1 + \beta : \beta \in S_0\} \\ & \cup \{\omega^{d-1}a_3 + \beta : \beta \in S_1\} \\ & \cup \{\omega^{d-1}a_5 + \beta : \beta \in S_2\} \\ & \cup \ \cdots \\ = & \bigcup_{i=0}^{\omega} H_i, \text{ where } H_i = \{\omega^{d-1}a_{2i+1} + \beta : \beta \in S_i\}, \end{aligned}$$

ordered by ordinal comparison. Every element of H_i is less than every element of H_{i+1}. Also, each H_i is order-isomorphic to S_i and thus ω^{d-1}. Therefore, $H \approx \omega^d$.

For any edge e in $\binom{H}{n}$, index its variables to satisfy criterion 1 of Definition 0.10 (this is possible because all low-exponent coefficients are distinct in H). Because we constructed the ω rows of H out of S_i with distinct coefficents, criterion 2 is satisfied. Criterion 3 is satisfied inductively for variables with exponents lower than $d - 1$. Because $\min A_i = a_{2^i}$ for all i and $2i - 1 < 2^i$ for all naturals $i \geq 1$, $a_{2i-1} < a$ for all $a \in A_i$. So criterion 3 is satisfied by e. Because A_0 is disjoint from all A_i with $i \geq 1$, criterion 4 is satisfied for variables with exponent $d - 1$, and by induction, it is satisfied for lower exponents. Because A_i is disjoint with A_j for all $i \neq j$, elements that differ in variables with exponent $d - 1$ differ in all lower-exponent variables. The induction with the previous statement satisfies criterion 5. Therefore e satisfies a CR of $\binom{\omega^d}{n}$. The coefficients in e are contained in G by the construction of H from G. □

Theorem 1.3. *For $n, d \in \mathbb{N}$, $T\left(n, \omega^d\right) \leq P\left(n, \omega^d\right)$.*

Proof. Let $c \in \mathbb{N}$ and let

$$\mathrm{COL}\colon \binom{\omega^d}{n} \to [c]$$

be a c-coloring of $\binom{\omega^d}{n}$.

Enumerate the CRs of $\binom{\omega^d}{n}$ as τ_0 to $\tau_{P(n,\omega^d)-1}$. The maximum size of any CR of $\binom{\omega^d}{n}$ is $n \cdot d$. For each τ_i, let

$$f_i\colon \binom{\omega}{n \cdot d} \to \binom{\omega^d}{n}$$

where if τ_i has size p, f_i maps X to the unique $e \in \binom{\omega^d}{n}$ where e satisfies τ_i and the p equivalence classes of e are made up of the p least elements of X. For example, one CR of $\binom{\omega^2}{2}$ is

$$a_{1,1} = a_{2,1} < a_{1,0} < a_{2,0}.$$

The corresponding f_i would be $f_i\colon \binom{\omega}{4} \to \binom{\omega^2}{2}$ with

$$f_i(x_1, x_2, x_3, x_4) = \{\omega \cdot x_1 + x_2, \omega \cdot x_1 + x_3\}$$

where $x_1 < x_2 < x_3 < x_4$. Note that f_i does not depend on x_4 – this is because the example CR has size 3, but the largest CR for $\binom{\omega^2}{2}$ has size 4.

Then, define $\mathrm{COL}'\colon \binom{\omega}{n \cdot d} \to [c]^{P(n,\omega^d)}$ with

$$\mathrm{COL}'(X) = (\mathrm{COL}(f_0(X)), \mathrm{COL}(f_1(X)), \ldots, \mathrm{COL}(f_{P(n,\omega^d)-1}(X)))$$

and apply Theorem 0.7 to find some $G \approx \omega$ where

$$\left|\mathrm{COL}'\left(\binom{G}{n \cdot d}\right)\right| = 1.$$

Let the one color in $\mathrm{COL}'\left(\binom{G}{n \cdot d}\right)$ be B. Note that B is a tuple of $P(n, \omega^d)$ colors.

Apply Lemma 1.2 to find some $H \approx \omega^d$ with the properties listed in Lemma 1.2. Now we claim

$$\left|\mathrm{COL}\left(\binom{H}{n}\right)\right| \leq P(n, \omega^d).$$

By Lemma 1.2, each element $e \in \binom{H}{n}$ satisfies a CR of $\binom{\omega^d}{n}$. Then for any arbitrary edge e, let e satisfy τ_i of size $p \leq n \cdot d$. Then take the p unique values in e, and if necessary, insert any new larger naturals from G to form a set of $n \cdot d$ values; denote this $A \in \binom{G}{n \cdot d}$. Then $\mathrm{COL}'(A) = B$ so by the definition of COL', $\mathrm{COL}(e) \in B$. Because $|B| = P(n, \omega^d)$, $T(n, \omega^d) \leq P(n, \omega^d)$. □

Proof. If $P(n, \omega^d) = 0$, then this is satisfied vacuously because $T(n, \omega^d) \geq 0$. Now suppose $P(n, \omega^d) \geq 1$. Note that all CRs of $\binom{\omega^d}{n}$ are disjoint from each other. That is, for any edge $e \in \binom{\omega^d}{n}$, if e satisfies τ', then it does not satisfy any

nonequivalent CR of $\binom{\omega^d}{n}$, where equivalence between CRs is defined by their $\overline{B}$ maps and $\preceq_{\overline{A}}$ orders being the same. This is because if e were to satisfy two CRs τ_1 and τ_2, then τ_1 and τ_2 must share the same equivalence classes and order, so the CRs must be equivalent. Therefore, we can index them $\tau_0, \ldots, \tau_{P(n,\omega^d)-1}$ and construct a coloring $\mathrm{COL}\colon \binom{\omega^d}{n} \to [P(n,\omega^d)]$ with

$$\mathrm{COL}(e) = \begin{cases} i & e \text{ satisfies } \tau_i \\ 0 & \text{otherwise} \end{cases}$$

Similar to the proof of Theorem 0.9, our coloring has two ways to output color 0, both through the satisfaction of τ_0 and through the catch-all case. The part that forces color 0 to be present in all order-equivalent subsets is the satisfaction of τ_0.

We now prove there is no ω^d-$(P(n,\omega^d)-1)$-homogeneous set. For all $H \approx \omega^d$ and for every CR τ of $\binom{\omega^d}{n}$, we find some $e \in \binom{H}{n}$ that satisfies τ. For arbitrary $H \approx \omega^d$ and τ, we find $e_{i,j}$ where

$$\{\omega^{d-1}e_{1,d-1} + \cdots + \omega^1 e_{1,1} + e_{1,0}, \ldots, \omega^{d-1}e_{n,d-1} + \cdots + \omega^1 e_{n,1} + e_{n,0}\}$$

satisfies τ.

We do this by assigning values to each $e_{i,j}$ according to where the equivalence class that contains $a_{i,j}$ is found in τ, moving left to right in the permutation of τ. By criterion 3 of Definition 0.10, each $e_{i,j}$ is assigned before $e_{i,j-1}$. As we do this, we ensure that if the leftmost unassigned value in τ is $e_{i,j}$, then

$$\{\omega^{d-1}e_{i,d-1}+\cdots+\omega^{j+1}e_{i,j+1}+\omega^j c_j+\omega^{j-1}c_{j-1}+\cdots+\omega^1 c_1+c_0 : c_i \in \omega\}\cap H \approx \omega^{j+1}.$$

By criterion 3 of Definition 0.10, the leftmost variable in τ must be $a_{i,d-1}$. Before any values are assigned, it is clear that

$$\{\omega^{d-1}c_{d-1} + \cdots + \omega^1 c_1 + c_0 : c_i \in \omega\} = \omega^d,$$

and because $H \subseteq \omega^d$, $\omega^d \cap H = H \approx \omega^d$.

By criterion 4 of Definition 0.10, all variables in an equivalence class must have the same exponent d. Let the leftmost unassigned equivalence class in τ be $a_{i_1,j} = a_{i_2,j} = \cdots = a_{i_m,j}$. By criterion 3, each $a_{i_i,\ell}$ for $1 \le i \le m$ and $\ell > j$ appeared to the left of this equivalence class and has already been assigned a value, and by criterion 5 the values for each exponent are equal: for all $\ell > j$ and $1 \le i \le m$, $e_{i_i,\ell} = e_{i_1,\ell}$.

By our previous steps,

$$\{\omega^{d-1}e_{i_1,d-1}+\cdots+\omega^{j+1}e_{i_1,j+1}+\omega^j c_j+\omega^{j-1}c_{j-1}+\cdots+\omega^1 c_1+c_0 : c_i \in \omega\}\cap H \approx \omega^{j+1}.$$

Then there exists some value e' where

$$\{\omega^{d-1}e_{i_1,d-1}+\cdots+\omega^{j+1}e_{i_1,j+1}+\omega^j e'+\omega^{j-1}c_{j-1}+\cdots+\omega^1 c_1+c_0 : c_i \in \omega\}\cap H \approx \omega^j,$$

where e' is greater than all previously assigned (and therefore finite) $e_{i,j}$ values. Then for $1 \le i \le m$, assign $e_{i_i,j}$ to be e'.

We can repeat this process to find $e_{i,j}$ for each CR of $\binom{\omega^d}{n}$ for arbitrary $H \approx \omega^d$. Therefore for all $H \approx \omega^d$, $\left|\mathrm{COL}\left(\binom{H}{n}\right)\right| \ge P(n,\omega^d)$ so $T(n,\omega^d) \ge P\left(n,\omega^d\right)$. With Theorem 1.3, we have $T(n,\omega^d) = P\left(n,\omega^d\right)$.

□

References

[1] A. Blass, N. Dobrinen, and D. Raghavan. The next best thing to a P-point. *The Journal of Symbolic Logic*, 80(3):866–900, 2015.

[2] A. S. Kechris, V. G. Pestov, and S. Todorcevic. Fraïssé limits, Ramsey theory, and topological dynamics of automorphism groups. *Geometric & Functional Analysis GAFA*, 15(1):106–189, Feb 2005.

[3] F. Ramsey. On a problem of formal logic. *Proceedings of the London Math Society*, 30(1):264–286, 1930.